AUTOSAR High-Level Design (HLD) Document

System Module: Electronic Braking & Stability System (EBSS)

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1. Document Control & Revision History

Version	Date	Author	Status	Summary of Changes
1.0	2026-01-15	System Architecture Team	Released	Initial baseline architecture for EBSS.
1.1	2026-02-20	Safety Integration Lead	Approved	Added failsafe redundancy metrics for pedal stroke sensors.
2.0	2026-03-10	Controls Engineering Group	Approved	Integrated Electronic Stability Control SWC and CAN message mappings.
2.1	2026-04-05	AUTOSAR RTE Team	Current	Refined client-server interfaces for MCAL valve block actuation.

2. Executive Summary & Architectural Scope

The Electronic Braking & Stability System (EBSS) is an AUTOSAR Classic Release 4.4 compliant chassis domain subsystem. The primary objective of EBSS is to execute deterministic, real-time deceleration and yaw stabilization for light passenger vehicles. It abstracts sensor data, evaluates wheel slip ratios, integrates driver braking demands, and dispatches modulated hydraulic pressure commands to the electro-hydraulic brake actuator block.

The software architecture strictly adheres to layered AUTOSAR separation: the Application Layer comprises dedicated Software Components (SWCs), which interface seamlessly with the Runtime Environment (RTE) and communicate across Basic Software (BSW) stacks including MCAL, ComStack, and the OS.

3. System Architecture & AUTOSAR Layer Overview

The EBSS architecture is structured into three fundamental AUTOSAR operational tiers: the Application Layer, the Runtime Environment (RTE), and the Basic Software (BSW).

3.1 AUTOSAR Layer Decomposition

Layer	Key Modules / Entities	Primary Function & Execution Model
Application Layer	BrakePedalSensorSWC AntiLockBrakingSWC ElectronicStabilityControlSWC BrakeActuatorDriverSWC	Implements core braking algorithms, slip estimation, and driver intent calculation. D
Runtime Environment (RTE)	Rte_EBSS_Generated_v2.1	Enables inter-SWC Sender-Receiver and Client-Server communication. Manages
Base Software (BSW) Services	Dem (Diag Event Mgr) FiM (Func Inhibit Mgr) NvM (Non-Volatile Mem)	Handles fault recording, safety degradation inhibit conditions, and persistent calibr
Communication Stack	CanTp, CanIf, PduR, Com	Marshals cyclic CAN frames (e.g. CAN_MSG_BRAKE_STATUS) over vehicle pow
MCAL Drivers	Adc, Pwm, Spi, Port, Dio	Direct hardware abstraction for dual pedal potentiometer ADC channels and solen

3.2 Hardware-to-Software Abstraction

Sensor signals enter the microcontroller via the Microcontroller Abstraction Layer (MCAL) ADC peripheral driver. The Sensor-Actuator SWCs package these physical ADC voltages into engineering units (% pedal travel, bar hydraulic pressure) and expose them across standardized Sender-Receiver interfaces to the core algorithmic Application SWCs.

4. Software Component (SWC) Specifications

This section details the functional responsibilities, internal runnable entities, and operational boundaries of each Application SWC.

4.1 BrakePedalSensorSWC (Sensor-Actuator Component)

Component Identifier: SWC_BrakePedalSensor
Execution Period: 5 ms cyclic runnable (Task_BrakeSensor_5ms)
Responsibilities:

- Reads raw dual potentiometer ADC voltages from channels ADC_CH_PEDAL_1 and ADC_CH_PEDAL_2 via MCAL.
- Computes cross-channel plausibility: if discrepancy between sensor 1 and sensor 2 exceeds 5.0% for > 40 ms, raises fault diagnostic DTC_PEDAL_PLAUSIBILITY_ERR.
- Converts verified analog stroke readings into normalized percentage signal Sig_PedalTravel_pct (0.0% to 100.0%).
- Categorizes driver deceleration urgency into Sig_BrakeIntent_enum (COMFORT_BRAKING, PANIC_BRAKING, STANDSTILL_HOLD).

4.2 AntiLockBrakingSWC (Algorithm SWC)

Component Identifier: SWC_AntiLockBraking
Execution Period: 10 ms cyclic runnable (Task_ABS_10ms)
Responsibilities:

- Calculates longitudinal slip ratio for each individual wheel: FL, FR, RL, RR based on wheel speed pulses Sig_WheelSpeed_XX_kmh and vehicle reference speed.
- Detects incipient wheel lockup when wheel deceleration gradient exceeds -2.5g and slip ratio exceeds 18%.
- Triggers three-phase hydraulic modulation cycle (Pressure Hold, Pressure Decay, Pressure Re-apply) to maintain maximum tire-road friction.
- Publishes ABS active status flag Sig_ABS_Active_flg to the vehicle chassis coordinator over the RTE.

4.3 ElectronicStabilityControlSWC (Algorithm SWC)

Component Identifier: SWC_ElectronicStabilityControl
Execution Period: 20 ms cyclic runnable (Task_ESC_20ms)
Responsibilities:

- Monitors lateral vehicle dynamics via inertial measurement unit signals: yaw rate Sig_YawRate_deg_s and lateral acceleration Sig_LateralAccel_g.
- Compares vehicle yaw response against standard Bicycle Dynamics Model derived from steering wheel angle.
- Mitigates understeer by applying targeted differential braking to inside rear wheel; mitigates oversteer by braking outside front wheel.
- Negotiates priority torque arbitration with Engine Management SWC to request powertrain throttle roll-back during severe skids.

4.4 BrakeActuatorDriverSWC (Actuator Component)

Component Identifier: SWC_BrakeActuatorDriver
Execution Period: 5 ms cyclic runnable (Task_Actuator_5ms)
Responsibilities:

- Translates target hydraulic pressure demands (Sig_TargetPressure_FL_bar to RR) into calibrated duty cycles for inlet/outlet solenoid valves.
- Controls hydraulic pump motor brushless motor controller through SPI commands sent to MCAL Spi driver.
- Performs continuous coil resistance monitoring to detect open-circuit and short-to-ground conditions across all 8 hydraulic valves.

5. Interface Catalog & Signal Dictionary

Inter-SWC communication is strictly managed through the RTE using either Sender-Receiver (S/R) data ports or Client-Server (C/S) synchronous service operations.

5.1 AUTOSAR Interface Catalog

Interface Name	Pattern	Source SWC	Target SWC(s)	Description
If_DriverBrakeRequest	Sender-Receiver	BrakePedalSensorSWC	AntiLockBrakingSWC ESC SWC	Transmits calibrated pedal travel (%) and driver br
If_WheelSpeedData	Sender-Receiver	MCAL WheelSpeedIO	AntiLockBrakingSWC ESC SWC	Delivers 4-wheel independent velocity signals.
If_StabilityIntervention	Sender-Receiver	ESC SWC	AntiLockBrakingSWC EngineManagementSWC	Dispatches differential torque/pressure corrections
If_BrakePressureCommand	Sender-Receiver	AntiLockBrakingSWC	BrakeActuatorDriverSWC	Transmits target hydraulic pressures per caliper (b
If_ActuatorHealthService	Client-Server	BrakeActuatorDriverSWC	DiagnosticEventMgr	Synchronous diagnostic routine to query valve coi

5.2 Signal Dictionary

Signal Name	Data Type	Range / Units	Interface	Update Rate
Sig_PedalTravel_pct	float32	0.0 .. 100.0 %	If_DriverBrakeRequest	5 ms
Sig_BrakeIntent_enum	uint8	0=NONE, 1=COMFORT, 2=PANIC	If_DriverBrakeRequest	5 ms
Sig_WheelSpeed_FL_kmh	float32	0.0 .. 350.0 km/h	If_WheelSpeedData	10 ms
Sig_WheelSpeed_FR_kmh	float32	0.0 .. 350.0 km/h	If_WheelSpeedData	10 ms
Sig_WheelSpeed_RL_kmh	float32	0.0 .. 350.0 km/h	If_WheelSpeedData	10 ms
Sig_WheelSpeed_RR_kmh	float32	0.0 .. 350.0 km/h	If_WheelSpeedData	10 ms
Sig_TargetPressure_FL_bar	float32	0.0 .. 250.0 bar	If_BrakePressureCommand	5 ms
Sig_TargetPressure_FR_bar	float32	0.0 .. 250.0 bar	If_BrakePressureCommand	5 ms
Sig_ABS_Active_flg	boolean	0=Inactive, 1=Active	If_StabilityIntervention	10 ms
Sig_YawRate_deg_s	float32	-120.0 .. +120.0 deg/s	If_StabilityIntervention	20 ms
Sig_LateralAccel_g	float32	-3.0 .. +3.0 g	If_StabilityIntervention	20 ms

5.3 CAN Bus Frame Packaging

The external status frame **CAN_MSG_BRAKE_STATUS** is mapped to CAN ID **0x1A4** and broadcast cyclically every **10 ms** across the vehicle High-Speed CAN chassis bus. Payload includes Sig_ABS_Active_flg (bit 0), Sig_PedalTravel_pct (bytes 1-2, scaled x0.1), and primary system status bits.

6. Inter-Component Dependencies & Timing Constraints

To ensure deterministic hard real-time execution in accordance with ISO 26262 ASIL-D requirements, strict data flow precedence and task execution cycles are enforced.

6.1 Data Flow & SWC Interaction Hierarchy

1. **BrakePedalSensorSWC** executes first in the 5 ms cycle, sampling sensor inputs and updating the RTE buffer.
2. **AntiLockBrakingSWC** depends directly on the output of BrakePedalSensorSWC and MCAL WheelSpeedIO. ABS calculations cannot proceed without fresh wheel speeds.
3. **ElectronicStabilityControlSWC** arbitrates with AntiLockBrakingSWC. When ESC triggers an active yaw correction, its wheel braking demand supercedes standard ABS slip control on the stabilizing wheel.
4. **BrakeActuatorDriverSWC** acts as the terminal sink component, receiving target pressures from ABS and ESC and commanding the hardware.

6.2 Fault Handling & Safe States

- **Dual Pedal Sensor Plausibility Failure:** If both stroke potentiometers disagree by > 5% for 40 ms, the system enters DEGRADED_PEDAL_FALLBACK mode. The regenerative braking request is cancelled and hydraulic backup is mandated.
- **Wheel Speed Loss:** If one wheel speed signal drops out, ABS function for that specific wheel is suppressed while retaining 3-wheel slip modulation.
- **Hydraulic Valve Driver Over-Temperature:** If actuator thermal sensor exceeds 135°C, high-frequency ABS pulsations are curtailed to protect MOSFET bridge drivers.
- **Complete System Failsafe State:** Upon critical RTE communication timeout (> 50 ms), the system cuts all solenoid power, causing valves to revert to mechanical fail-safe pass-through mode.

6.3 What is NOT Covered in this Document (Out of Scope)

This specification intentionally excludes: Long-range radar adaptive cruise control algorithms, electronic parking brake (EPB) cable puller actuators, high-voltage traction inverter regenerative blending, and steer-by-wire pinion actuators. These functions reside in separate chassis and ADAS HLD documents.